

LESSON GUIDE

Technology Plus

Unit 3: Hardware

2.3.2 – Network Storage Devices

Lesson Overview:

In this lesson, students will identify and compare local network storage devices.

Students will:

- Identify network storage and its purpose.
- Describe common types of network storage.
- Distinguish between network storage and storage drives.

Guiding Question: What is network storage, its purpose, and how is it different than storage drives?

Suggested Grade Levels: 8-12

Technology Needed: Yes

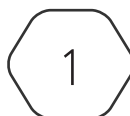
Approximate Time Required: 65-85 minutes

Standards:

CompTIA Tech+ FC0-U71 Objective

- 2.3 – Compare and contrast storage types.
 - o Volatile vs. non-volatile
 - o Local storage
 - RAM
 - Read-only memory (ROM)
 - Storage drive
 - Magnetic disks/hard disk drive (HDD)
 - Solid-state drive (SSD)
 - Non-volatile memory express (NVMe)
 - Optical
 - External flash drives
 - o Local network storage
 - Network-attached storage (NAS)
 - File server
 - Cloud storage service

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Lesson Background

Background Information:

In this lesson, students will explore network storage devices and how they compare to local storage drives. Students will be able to recognize that local network storage provides access to data for anyone that is on the network while storage drives can only access data on a specific device. Network-attached storage and file servers are specific devices that are utilized for storing files and sharing them over a network. Cloud storage is network storage available online over remote servers.

Students will practice sharing, accessing, and editing files over a network using the Apache2 web server and SSH. By the end of the lesson students should be able to identify common types of network storage and distinguish them from storage drives.

Materials Needed:

Materials per Class	• Lesson Guide • Lesson Slides 2.3.2 – Network Storage Devices • Lab Slides 2.3.2 – Network File Access with SSH
Materials per Student	• Student Notes 2.3.2 – Network Storage Devices • Student Handout 2.3.2 – Network Storage Devices • Assessment 2.3.2 – Network Storage Devices (optional)

Cyber Terms:

Term	Definition
local network storage	storage devices connected to a local network within a specific location
network-attached storage (NAS)	a device connected to a network that provides file-based data storage services to other devices on a network
local area network (LAN)	a small network in one location, like a home, classroom, or office
file server	a server dedicated to managing and storing files and providing file access to other computers or devices on a network
cloud storage	an online service that keeps data on remote servers that are accessible over the internet

2.3.2 – Network Storage Devices

Guiding Question: What is network storage, its purpose, and how is it different than storage drives?

Lesson Launch (10-15 minutes)

Scenario: Local Network Storage

- At your school, you log into a computer and save a document to the “Shared Student Drive”. Later, when you are in a different classroom, you log in again and the same file is still there.
- Even though you are using a different computer, you are still able to access the file because it is stored on your account within the school’s local network, a central place where all student’s files are saved and managed.
- Local network storage allows you to log into any computer at school and still access your files. You don’t need to carry around a flash drive or email yourself every time a file is altered.

Local Network Storage vs. Local Storage Drives

- Storage connected to a local network is shared with multiple devices over that network.
- Data is available to anyone on the network, with permission.
- Local network storage is like a binder in a teacher’s cabinet that everyone in the class can access.
- Storage drives are directly attached to a single device.
- Only a specific device can access the data unless it is manually shared.
- Local storage drives are like a personal notebook in your backpack.

Beyond the USB: Network Storage (20-25 minutes)

- Distribute Student Handout 2.3.2 – Network Storage Devices and encourage students to fill in the answers during the lesson.

Local Network Storage

- Introduce **local network storage** and discuss its advantages and disadvantages.
 - The two common local network storage options we will discuss are network-attached storage and file servers.
- Introduce a **network-attached storage (NAS)** device, its characteristics, advantages and disadvantages.
 - Advantages:
 - Usually includes redundant array of disks (RAID) configurations for redundancy and data protection.
 - Redundancy refers to storing multiple copies of the same data in different locations.
 - Can be configured for remote access via the internet.

- Disadvantages:
 - Limited by the local network's bandwidth and speed.
 - High initial cost and ongoing maintenance requirements.
- Introduce a **file server**, its characteristics, advantages and disadvantages.
 - Server message blocks (SMBs) are common in Windows-based environments.
 - Network file systems (NFSs) are common in Linux/Unix-based environments.
 - Advantages:
 - Supports complex file-sharing needs – including permissions and user access controls.
 - Can be scaled and configured based on network needs.
 - Disadvantages:
 - If the server goes down, users can lose access to all shared files.
 - Power outage, network failure, hardware issue, OS crash or freeze, software update error can cause the server to go down.
 - Requires dedicated hardware and software resources.
- Introduce **cloud storage** service, its characteristics, advantages and disadvantages.
 - Advantages:
 - Provides remote access to data from multiple locations and devices.
 - Storage capacity can be instantly adjusted as needed.
 - Disadvantages:
 - Requires an internet connection for access.
 - Requires an ongoing subscription which can be costly.

Lab: Network File Access with SSH (30-40 minutes)

- This lab requires access to the CYBER.ORG Range – you will need to launch the Ubuntu, Kali, and Windows Server virtual machines – and Lab Slides 2.3.2 – Network File Access with SSH.
- Introduce the guiding question and objectives for the lab.
- In this lab students the Kali VM will act as a “file” server while the Ubuntu VM and Windows Server VM act as clients. Students will practice sharing, accessing, and editing files over a local network using the Apache2 web server.
- Discuss lab key terms.
- Students should open three separate tabs in their browser for each virtual machine. This will allow them to easily switch back and forth during the lab.
- Students should complete the lab version of their student handout as they work through the lab.

Helpful Tips

- Remind students to type carefully when they are typing in the terminal.

- Students can use the up arrow on the keyboard to roll back previous commands.
- When the slide uses angle brackets (< >) in a command, students should replace what is inside the brackets with their text. i.e. the name of the file or their Kali IP address.
- Apache2 is free and open-source web server software.
- Depending on the file type, when accessing through the browser – some files will load into a window, and some will download to the computer. The files students are working with today are .txt and .html – they will open in a window. If they were working with .exe or .zip they would download to the computer.
- Students can choose two files they want to load in the browser vs. using CLI. Students should access a .html and a .txt file for each method.
- SSH, or secure shell network protocol, is designed for secure communication over an unsecured network. Key features of SSH are encryption, authentication, data integrity, and tunneling.
- If we do not start SSH and edit the SSH configuration file for password and pubkey authentication – the client machines will not be able to connect to the server machine via SSH.
- In this lab SSH, authenticates the client's access to the server with the username, IP address, and password. SSH also uses encryption and integrity to ensure that our commands and file edits cannot be changed by a possible sniffer on the network.
- Without sudo students would not be able to edit the files because permission to write is only accessible to the root (administrative) user.
- Because we are accessing Kali through SSH in the client machines, every command we type is executed in Kali Linux terminal.
- When editing the .html file in Windows, students should leave the html tag (<h1>) and only edit the text inside.
- Students should understand that editing a file in both clients would look the same, i.e. editing a .html in Ubuntu will look the same as it did in Windows.
- Students should choose the method they want to check that files were updated using SSH. Encourage students to use a different method for each machine, i.e. browser for Ubuntu and CLI for Windows.

Put It All Together

- Did you notice any differences when using Windows as a client versus Ubuntu?
We used different commands in the CLI to access the files from the Kali machine. In Windows we used **curl** and in Ubuntu we used **wget**. In Ubuntu we used the **ls** command to view files in a directory and in Windows we used **dir**. In Windows to open and view the contents of a file we used the command **start <file_name>** and in Ubuntu we use **xdg-open <file_name>**.
- Did you notice any similarities when using Windows as a client versus Ubuntu?
Both machines allow access to files in the Apache2 directory through a browser search bar. Both machines used the same command to access Kali using SSH.
- What was the purpose of starting and configuring SSH in the Kali Linux machine?

So we can connect our client machines to Kali successfully using its username, password, and IP address.

- How did we know the lab was successful?

We knew the lab was successful when the changes we made on the server machine updated in real time and were visible from the client when viewing the files via the browser or CLI.

Putting It All Together (5 minutes)

- Review what students learned today by discussing the questions below as a class.
- What is the difference between local storage drives and local network storage devices?

A local network storage consists of storage devices connected to a local network within a specific physical location. Anyone on the network can access any data stored on a device that is connected to the network.

- How does local network storage differ from cloud storage?

Cloud storage keeps data offsite and is accessible over the internet from any remote location. Local network storage keeps data within a specific physical location and requires maintenance by onsite IT professionals.

- How can a user distinguish between a file server and network-attached storage?

File servers are servers that use networking protocols like SMB and NFS for file sharing and can be easily scaled or configured based on network needs. NAS devices are devices that connect to the LAN using a wired Ethernet connection and come with their own OS. File servers centralize files while NAS devices use multiple hard drives for redundancy.